

ASIFA AFZAL

Software Engineer | Backend Developer

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OBJECTIVE

Motivated and detail-oriented Computer Science graduate with a strong foundation in software development, data analysis, and backend technologies. Skilled in Python programming along with frameworks such as Django and FastAPI, with hands-on experience in building web-based and data-driven applications. Proficient in working with relational databases including MySQL, PostgreSQL, and SQLite, along with practical exposure to data analysis tools like Pandas, NumPy, Seaborn, Matplotlib and visualization libraries. Familiar with containerization using Docker, CI/CD workflows via GitHub Actions, and basic cloud deployment on AWS services. Eager to contribute to scalable, efficient, and user-focused backend solutions while continuously enhancing skills in software engineering, cloud technologies, and AI-driven systems.

EDUCATION

Bachelor of Science in Information Technology (BS IT)

Sep 2024 – Jan 2026

Virtual University of Pakistan | CGPA: 3.6 / 4.0

Relevant Coursework & Academic Focus:

- Programming Fundamentals
- Object-Oriented Programming
- Database Management Systems (SQL)
- Web Development (Frontend & Backend Basics)
- Artificial Intelligence & Machine Learning (Introductory)
- Data Structures & Algorithms

Bachelor of Science in Computer Science (BSC) / Associate Degree in Science

Oct 2021 – Nov 2022

Mathematics | Computer Science | Economics — Independent College, 179 Jinnah Colony, Faisalabad

Intermediate (Computer Science)

Aug 2019 – Oct 2020

Mathematics | Computer Science | Economics — Independent College, 179 Jinnah Colony, Faisalabad

EXPERIENCE

Python Developer — Syntecxhub

Internship

- Built and maintained backend features and scripts using Python, working with data structures, APIs, and database-driven logic.
- Collaborated with the team on debugging, testing, and improving code quality for ongoing projects.

Django Developer — Hail Technologies

Internship

- Developed and maintained web application modules using the Django framework, including models, views, and REST endpoints.
- Worked with relational databases (MySQL/PostgreSQL) to design schemas and implement CRUD functionality for application features.

PROJECTS

Full-Stack Design — SecureVault

Designed a Python and Django-based, YouTube-style, password-protected video sharing platform with a four-tier role hierarchy (CEO, Manager, Admin, User/Viewer) and approval-based access control. Every uploaded video is auto password-protected, with role-based dashboards, an access request/approval workflow, and full UI/UX documentation across 10 core modules.

Design Scope: Python, Django, Role-Based Access Control (RBAC), System Architecture, UI/UX Documentation

Data Analyst — Portfolio Analytics Dashboard

Designed and developed a Portfolio Analytics Dashboard that converts raw portfolio data into meaningful insights through data analysis and visualization, providing a clear overview of project performance, technology usage, skill distribution, and portfolio trends.

- Portfolio Performance Metrics Analysis
- Technology Usage Breakdown
- Skill Category Distribution
- Project Status Monitoring

- Project Category Analysis
- Experience Level Insights
- Data-Driven Visualizations and Reporting

Tools & Technologies: Python | SQL | Excel | Data Visualization | Data Analysis

PHP-Based Management System — Local Trip Organizer

Developed a web-based trip management system to help users plan and organize local trips efficiently, allowing users to manage trip details, bookings, and destinations through an interactive interface.

Technologies: PHP, HTML, CSS, JavaScript, MySQL, WampServer64, Dreamweaver

Web-Based Skill Sharing System — Community Skills Exchange Platform

Developed a full-stack web application enabling users to exchange skills and services without monetary cost. The platform supports user registration and JWT-based authentication, personalized dashboards, skill listings with category-based search and filtering, a booking and exchange request system, email notifications, and a rating & review system. An admin panel was implemented for managing users, skill listings, categories, and abuse reports.

- Frontend: HTML5, TailwindCSS, JavaScript
- Backend: PHP (Server-side logic & REST APIs)
- Database: MySQL (via XAMPP)
- Security: JWT Authentication, bcrypt password hashing, protection against SQL Injection, XSS & CSRF
- Media Storage: Cloudinary (image upload & management)

Python Developer — Web Scraper Bot

Built a Python-based web scraping bot in PyCharm to automate extraction of data from websites, including request handling, HTML parsing, and structured data export for further analysis.

- Automated data extraction from target web pages using HTTP requests and HTML parsing
- Structured and exported scraped data for storage and downstream use
- Handled errors and basic rate-limiting to keep scraping stable and reliable

Technologies Used: Python, PyCharm, Requests, BeautifulSoup

TECHNICAL SKILLS

- Programming: Python, C++, PHP
- Databases & Caching: PostgreSQL, MySQL, SQLite, Redis (Caching & Message Broker)
- Frontend (Familiar): JavaScript (React), HTML5, CSS3, Bootstrap, REST API
- Languages & Frameworks: Python (Django, FastAPI), Django REST Framework, Kotlin
- DevOps & Tools: GitHub Actions (CI/CD), Git, VS Code, Jupyter Notebook, Google Colab, Kaggle
- Core Competencies: RESTful API Design, System Architecture, Algorithmic Problem Solving
- AI & Machine Learning: Prompt Engineering, NLP (Basic Understanding)
- Data Analysis: NumPy, Pandas, Matplotlib, Seaborn

CERTIFICATES

- C++ – CISCO Network
- Python – CISCO Network
- CCNA – CISCO Network
- Artificial Intelligence – NAVTCC
- Generative AI Application Developer – PAK-ANGELS
- Core Django – Skill Grid

LANGUAGES

- Urdu
- English